Andy Phu

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Education

University of California, Irvine Graduated Dec 2024
B.S. Computer Science | Concentration in Machine Learning *Irvine, CA*
Coursework: Distributed Systems, Machine Learning, Relational Databases, Data Structures, Algorithms

Experience

Uber Oct 2025 – Present
Software Engineer

- Integrated Autonomous Vehicle delivery partners into the Accesspoints stack, replacing low-value pickup/dropoff locations and reducing defective location error by **70%**, from 500m to 150m at p99
- Built a 3-stage AI agent pipeline to classify customer complaint tickets, generate automated debugging reports, and surface recurring issues, eliminating **140 hours** of manual engineering effort per month
- Engineered and validated three categorical features for an XGBoost pickup-location ranking model across **160M+** SQL training samples, reducing defective pickup errors by **20%** in production
- Architected a **5TB** Apache Spark ETL pipeline processing trip data to generate training data for the first offline Delivery Accesspoint recommendation model and Tier 1 metrics powering ML and A/B experimentation

GEICO Jan 2025 – Oct 2025
Software Engineer

- Developed scalable API endpoints by converting legacy C# services to Go, enabling seamless customer access to eligible coverages, leveraging Go, SQL, Azure, and Kubernetes to enhance system scalability and reliability
- Architected a Go comparison service module to ensure data consistency across legacy and modernized endpoints, which facilitated a seamless migration and improved data accuracy
- Led the migration to K6 load testing for critical endpoints processing 1 million requests daily, configured benchmarks, and used Grafana to monitor performance, ensuring reliability before production releases

BlackRock May 2024 – July 2024
Software Engineer Intern

- Designed and engineered a full stack application for the finance department to log and manage portfolios directly in the SQL database, facilitating self-service management without managerial oversight
- Collaborated with three team members to develop a modularized AI-powered assistant using the OpenAI API, streamlining the new hire onboarding process, with a projected savings of 600 hours for the firm
- Developed an ETL pipeline in Python to validate 5,000 weekly stock entries and automatically update the Snowflake database for stock purchases, reducing manual reconciliation efforts by 90%, which were both error-prone and time-consuming

Projects

OpenHouse (Interior Design Recommendation System) | *Go, TypeScript, Next.js, Pinecone, PostgreSQL, Prisma*

- Developed a full stack Tinder-style furniture discovery app using Go and Next.js, integrating web scraping and concurrent data processing to collect and analyze thousands of furniture listings
- Engineered a machine learning-powered recommendation system using Hugging Face vector embeddings and Pinecone vector search, ranking furniture dynamically based on user preferences and similarity scores

Technical Skills

Languages: Go, Java, Python, TypeScript, C++, JavaScript, C#, HTML, CSS, SQL, R, C
Tools/Frameworks: AWS, Docker, Kubernetes, gRPC, PostgreSQL, React.js, MongoDB, Express, TailwindCSS, Git, FastAPI, Node.js, Flask, Prisma, Google Cloud, Oracle Cloud, PyTorch, Pandas, Cassandra, Apache Spark, SnowFlake, Azure, Nginx, .NET, Redis, Grafana, Kafka, Presto, XGBoost, Cadence, Apache Airflow, Apache Hive, HDFS, YARN